

CHAPTER 1

INTRODUCTION

Aquaculture, an essential component for achieving global food sustainability, has gained increasing prominence as the world's population is projected to exceed 9.8 billion by 2050 [1]. Traditional harvesting from natural resources, especially the ocean, is currently beyond ecological limits and is incapable of meeting escalating food demand [2]. Thus, aquaculture emerges as an efficient method for protein production, contributing to food security while attempting to preserve marine biodiversity [3]. However, the environmental impact associated with aquaculture necessitates comprehensive ecosystem management to ensure sustainability.

Water quality monitoring (WQM) and water quality prediction (WQP) play a vital role in managing aquaculture, as water quality (WQ) is the most influential factor affecting both the production and product quality [4]. Factors such as fish density, quality of the feed, feeding interval, and climate, can affect WQ, potentially triggering problems like stress, reduced food intake, increased vulnerability to diseases, and higher fish mortality rates [5, 6]. Traditionally, WQM involves collecting water specimens for lab testing, a time-consuming process that can delay critical decision-making [7]. Furthermore, most available WQM systems, which often use costly probes, are unaffordable for small-scale farms that significantly contribute to overall production [8, 9]. Some research works have focused on reducing the cost of WQM systems by minimizing sensor costs [10–12], which has proved beneficial for farmers by improving aquaculture production and sustainability.

Recent advancements in technology have introduced smart aquaculture systems by integrating IoT, artificial intelligence, and data analytics [13, 14]. However, accurate WQP remains a challenge due to the non-linear characteristics of WQ parameters and their complex interdependencies [15]. Conventional WQP approaches with machine learning (ML) techniques such as autoregressive moving average model (ARMA), autoregressive integrated moving average (ARIMA), seasonal autoregressive integrated moving average model (SARIMA), seasonal auto-regressive integrated moving average with exogenous factors (SARIMAX), Holt-Winters exponential smoothing (HWES), Markov model, grey model, and support vector regression (SVR) have limitations. These models, assuming a linear relationship with

WQ parameter data, are less accurate and take longer prediction time, making them unsuitable for accurately predicting non-linear aquaculture WQ parameters [16–18].

Deep learning (DL) models like long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN) overcome these limitations and capture the non-linear nature of aquaculture WQP effectively [19]. These models have achieved success in various applications, making time series prediction more precise and efficient in terms of training time and required processing power [20].

In this context, this research explores various DL models and aims to propose innovative ones, specifically tailored for accurate, low-cost WQM and precise WQP in aquaculture. The focus of this research is to propose DL models which can enhance prediction accuracy, while concurrently reducing the computation time. This directly addresses the urgent necessity for efficient WQ management in aquaculture, facilitating improved performance and reliability of WQM as well as WQP systems. Additionally, these models aim to offer an economically viable solution, reducing the overall cost of the WQM system, thereby ensuring affordable access to advanced and technology assisted WQ management.

1.1 MOTIVATION AND MAJOR OBJECTIVES

The inspiration driving this research is the need for affordable, real-time, and accurate WQM and WQP systems in aquaculture. Traditional monitoring methods are not sustainable, and they need to be more efficient in dealing with the dynamic complexities of the ecosystem. In parallel, the field of DL, a subset of artificial intelligence, has demonstrated potential in managing complex, dynamic, and non-linear systems. It is thus compelling to explore and adapt these techniques to the unique challenges of WQ management in aquaculture.

The objectives of this research, reflecting the primary motivation, are structured as a series of interconnected works:

1. Development of a cost-effective virtual sensing (VS) system for real-time WQM, reducing dependence on costly physical sensors. This involves building a deep learning enhanced virtual sensing (DLEVS) system that harnesses the potential of DL to enhance the accuracy of VS-based WQM systems significantly.
2. Exploration of the potential of LSTM, GRU, and their bidirectional variants in implementing a WQP system, with an emphasis on improving prediction accuracy.
3. Further refinement of the accuracy and computational efficiency of WQP

through the proposal of hybrid DL models. This work extensively studies the impact of hyperparameters (h_p) in optimizing these models for peak performance.

4. Proposal of attention-based DL models to further enhance accuracy and computational efficiency of WQP. This work also includes an extensive study of the model performance with tuning of h_p .

Collectively, these objectives aim to integrate advanced DL methodologies into aquaculture WQ management, striving for improved accuracy, efficiency, and sustainability. The proposed research intends to develop and implement DL models that can accurately predict WQ parameters in real-time, thereby enhancing the sustainability of aquaculture operations. These models are designed to perform efficient and affordable WQM as well as accurate WQP.

1.2 CONTRIBUTIONS

The research conducted in this thesis makes several significant contributions to the field of aquaculture WQM and WQP using DL techniques. These major contributions are as follows:

- Development of a cost-effective virtual sensing (VS) system for real-time WQM: A deep learning enhanced virtual sensing (DLEVS) system is proposed to address the challenges of costly physical sensors in WQM. By harnessing the power of DL, the DLEVS system significantly enhances the accuracy of VS-based WQM, providing a cost-effective solution for real-time monitoring of WQ parameters.
- Implementation of LSTM, GRU, and their bidirectional variants for WQP: The thesis investigates the potential of these DL models for accurate WQP in aquaculture. By leveraging the non-linear modeling capabilities of LSTM and GRU, the models improve prediction accuracy, providing valuable insights into future WQ conditions.
- Refinement of WQP accuracy and computational efficiency through hybrid DL models: The research proposes hybrid CNN-LSTM/GRU models for WQP, aiming to optimize the performance of the models. Extensive studies on hyperparameters are conducted to enhance the accuracy and computational efficiency of the models, making them more practical for real-time implementation.
- Proposal of attention-based DL models for enhanced WQP: The thesis presents attention mechanisms into the DL models for further enhancing the prediction

accuracy and computational efficiency in WQP. By focusing on the most relevant features and parameters, the attention-based models improve the prediction performance, ensuring more accurate and efficient WQP.

Overall, the contributions of this research include the development of a cost-effective VS and DL based WQM system, exploration and implementation of DL models for accurate WQP, refinement of prediction accuracy and computational efficiency through hybrid models, proposal of attention-based models, and integration of advanced methodologies for WQ management. This research work integrates the DL models into aquaculture WQ management. These contributions provide valuable insights for the development of sustainable aquaculture practices through smart WQM and WQP.

1.3 THESIS ORGANIZATION

Chapter 1 introduces the thesis by presenting the motivations behind the research and outlining the main objectives aimed to achieve. This chapter also highlights the key contributions of the thesis. With the aforementioned research motivation and objectives, this thesis investigates the development of various approaches to address WQM and WQP in aquaculture using DL techniques. The subsequent chapters of the thesis are organized and presented as follows:

Chapter 2 provides a comprehensive literature review on WQM and WQP in aquaculture. The chapter categorizes existing algorithms and methodologies, discusses their limitations, and analyses performance measures commonly used in this field. Additionally, an overview of relevant databases and the performance metrics used in the research are provided.

Chapter 3 focuses on the development of a VS based WQM system using DL techniques. The proposed methodology, including VS and DL is explained in detail. The chapter describes the experimental setup, including the hardware and software used, and presents the results and discussions based on the ADAK and MAC datasets.

Chapter 4 introduces DL models for WQP in aquaculture. Specifically, the LSTM, GRU, and bidirectional LSTM/GRU DL models are proposed. The chapter presents the experimental setup, including details on the conducted experiments, and provides results and discussions based on the ADAK as well as MAC datasets.

Chapter 5 focuses on the development of hybrid CNN-LSTM/GRU models for WQP. The proposed models and their implementation details are presented in this chapter. The experimental setup, experiments conducted, and results and discussions

based on the ADAK as well as MAC datasets are provided.

Chapter 6 explores attention mechanism-based DL models for WQP in aquaculture. The proposed attention-based bidirectional recurrent neural network (ABD-RNN) models, which include the bidirectional LSTM/GRU DL layer and attention layer, are discussed in detail. The chapter presents the experimental setup, describes the conducted experiments, and provides results and discussions based on the ADAK and MAC datasets.

Chapter 7, summarizes the entire dissertation and provides a comprehensive overview of the contributions made to the research. It also discusses the limitations of the accomplished work and suggests potential future research directions.

By adhering to this organizational structure, the dissertation thoroughly examines DL based WQM and WQP systems for aquaculture. The chapters cover various aspects, including VS, DL based WQP models such as hybrid models, and attention-based models. The dissertation aims to contribute to aquaculture by providing accurate and efficient WQ management solutions.